

Software Engineering Lab 1

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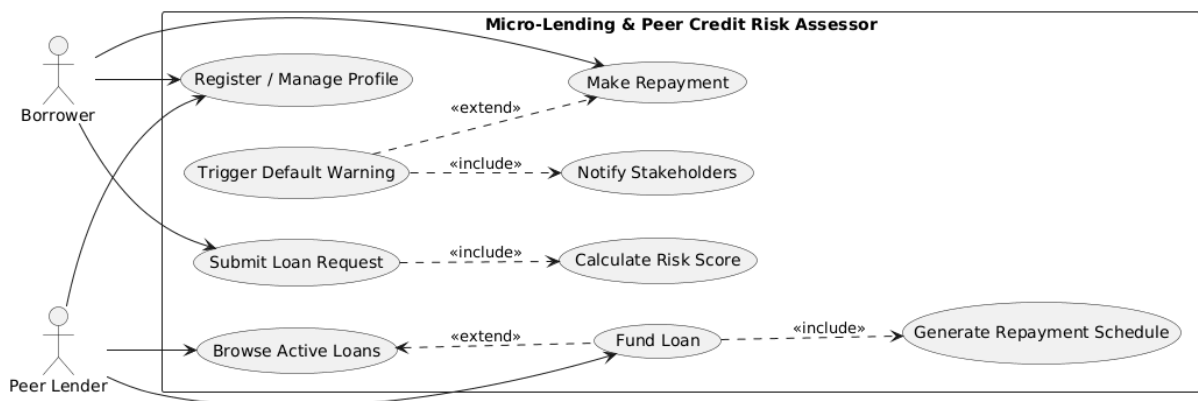
Deliverable 1: Requirements Table

ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall compute borrower credit risk ratings based on debt-to-income ratio, repayment history, and employment stability indicators.	High	Pass: Risk tier (Low/Med/High) accurately assigned; Fail: Incomplete financial profile generates credit approval.	Essential for peer lenders to gauge risk before funding.
FR-002	Functional	The system shall generate structured repayment schedules	High	Pass: Accurate EMI schedule with dates is generated	Borrowers need clear timelines to prevent defaults.

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		for funded loans.		upon loan approval.	
FR-003	Functional	The system shall trigger automated default warnings to borrowers and lenders when an EMI is missed.	Med	Pass: Warning notification sent at exactly T+1 days past the due date.	Proactive risk management and communication.
FR-004	Functional	Borrowers shall be able to submit new loan requests specifying the required amount and purpose.	High	Pass: Loan request is successfully saved and queued for risk assessment.	Core platform functionality for initiating the lending cycle.
FR-005	Functional	Peer Lenders shall be able to	High	Pass: Lender funds are deducted and the loan is	Enables the peer-to-peer transaction

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		browse and fund active loan requests based on borrower risk scores.		marked as "Funded".	aspect of the platform.
NFR - 001	Performance & Security	Loan repayment transactions and EMI distribution must execute inside database ACID transactions to prevent double-spending.	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.	Prevents severe financial data corruption.
NFR - 002	Usability	The platform's risk assessment and repayment dashboard must load within 2 seconds.	Med	Pass: Automated load testing confirms page load times <= 2000ms.	Ensures a smooth user experience for stakeholders.

Deliverable 2: UML Use-Case Diagram



Deliverable 3: Use-Case Flow Specification

Use Case Name: Compute Borrower Credit Risk Rating

1. Use Case Description This use case describes the process by which the Micro-Lending platform evaluates a borrower's financial profile to calculate and assign a credit risk tier (Low, Medium, or High).

2. Actors

- **Primary Actor:** System (Risk Assessor Engine)
- **Secondary Actor:** Borrower

3. Preconditions

- The Borrower has successfully registered an account on the platform.
- The Borrower has initiated a new loan request and populated their complete financial profile, including income, active debts, and employment history.

4. Postconditions

- **Success:** The Borrower is assigned a verified Risk Tier (Low/Med/High), which is saved to the database and made visible to potential Peer Lenders.

- **Failure:** The risk assessment is aborted, no score is assigned, and the Borrower is prompted to complete their profile.

5. Main Success Scenario (Standard Flow)

1. The Borrower submits a loan request.
2. The system triggers the Risk Assessor Engine.
3. The system retrieves the borrower's submitted financial data: debt-to-income ratio, repayment history, and employment stability indicators.
4. The heuristic rules engine evaluates the retrieved indicators against the platform's risk baseline.
5. The system successfully calculates a numerical risk score.
6. The system maps the numerical score to the corresponding Risk Tier (Low, Med, or High).
7. The system updates the loan request status to "Assessed" and displays the tier to both the Borrower and Peer Lenders.

6. Alternate Flow 1 (Incomplete Financial Profile)

- *2a.* During step 3 of the Main Success Scenario, the system detects missing mandatory data (e.g., missing employment stability indicators).
- *2b.* The system halts the risk calculation process immediately.
- *2c.* The system generates an error notification: "Incomplete financial profile. Please update your employment history to proceed."
- *2d.* The credit approval fails, and the loan request remains in the "Draft" state until the Borrower updates the required fields.

Paste this into Word, insert your UML diagram image right between Deliverable 1 and Deliverable 3, and save it as your single PDF. Let me know when you are ready to generate the Jira data for Lab 2.